Generalized Additive Model for Council District VI (2023 - 2025 Sales)

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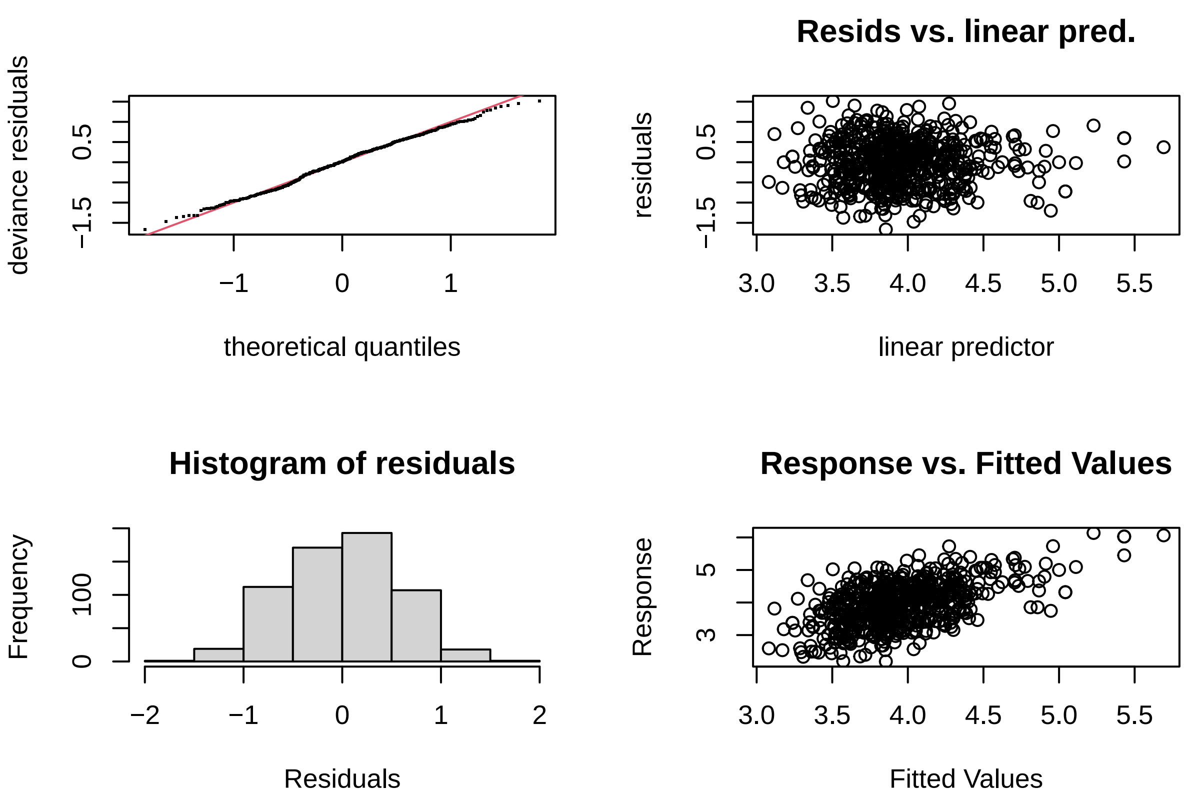
Contents

Model I

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_sppsf ~ arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + s(months,
##     bs = "cs", k = 10)
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)          3.884804   0.073730  52.690  < 2e-16 ***
## arterial             0.245048   0.081009   3.025 0.002595 **
## single_family1story  0.152873   0.063034   2.425 0.015596 *
## half_duplex1story    0.000000   0.000000     NaN      NaN
## half_duplex2story    0.078972   0.682939   0.116 0.907981
## smallest            -0.002988   0.095421  -0.031 0.975028
## small               -0.016897   0.072383  -0.233 0.815497
## large               -0.094109   0.071027  -1.325 0.185687
## largest              0.125176   0.133167   0.940 0.347604
## two_plus_baths       0.250168   0.153119   1.634 0.102829
## powder_room          0.236911   0.089159   2.657 0.008092 **
## good                 0.513812   0.145251   3.537 0.000436 ***
## fair                -0.275972   0.051697  -5.338 1.34e-07 ***
## poor                -0.395011   0.109623  -3.603 0.000341 ***
```

```
## very_poor_unsound   -0.266484   0.420898  -0.633 0.526892
## qabove_avg           0.561373   0.347244   1.617 0.106485
## qbelow_avg           0.029153   0.058167   0.501 0.616424
## crawl               -0.050340   0.081905  -0.615 0.539046
## slab                 0.104175   0.114825   0.907 0.364644
## fireplace           -0.250719   0.080418  -3.118 0.001911 **
## hot_water           -0.022968   0.063494  -0.362 0.717676
## elec_baseboard       0.000000   0.000000     NaN      NaN
## wall                 0.086993   0.159256   0.546 0.585104
## central_air          0.160246   0.090003   1.780 0.075513 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                        edf Ref.df     F  p-value
## s(sresb_yearbuilt) 4.3616      9 4.754  < 2e-16 ***
## s(ln_garagespaces) 0.8727      3 0.452   0.0732 .
## s(months)          2.4685      9 2.817 8.65e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Rank: 43/45
## R-sq.(adj) =  0.262   Deviance explained = 29.6%
## -REML =  573.5  Scale est. = 0.3324    n = 622
```




```
##
## Method: REML   Optimizer: outer newton
## full convergence after 6 iterations.
## Gradient range [-7.233609e-09,2.953726e-11]
## (score 573.4968 & scale 0.3324039).
```

```
## Hessian positive definite, eigenvalue range [0.1448696,299.0217].
## Model rank =  43 / 45
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                      k'   edf k-index p-value
## s(sresb_yearbuilt) 9.000 4.362    0.90   0.010 **
## s(ln_garagespaces) 3.000 0.873    0.91   0.015 *
## s(months)          9.000 2.468    0.91   0.030 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1   622     0.000608   0.000741    0.000478  1.55  57.8 -0.869
```

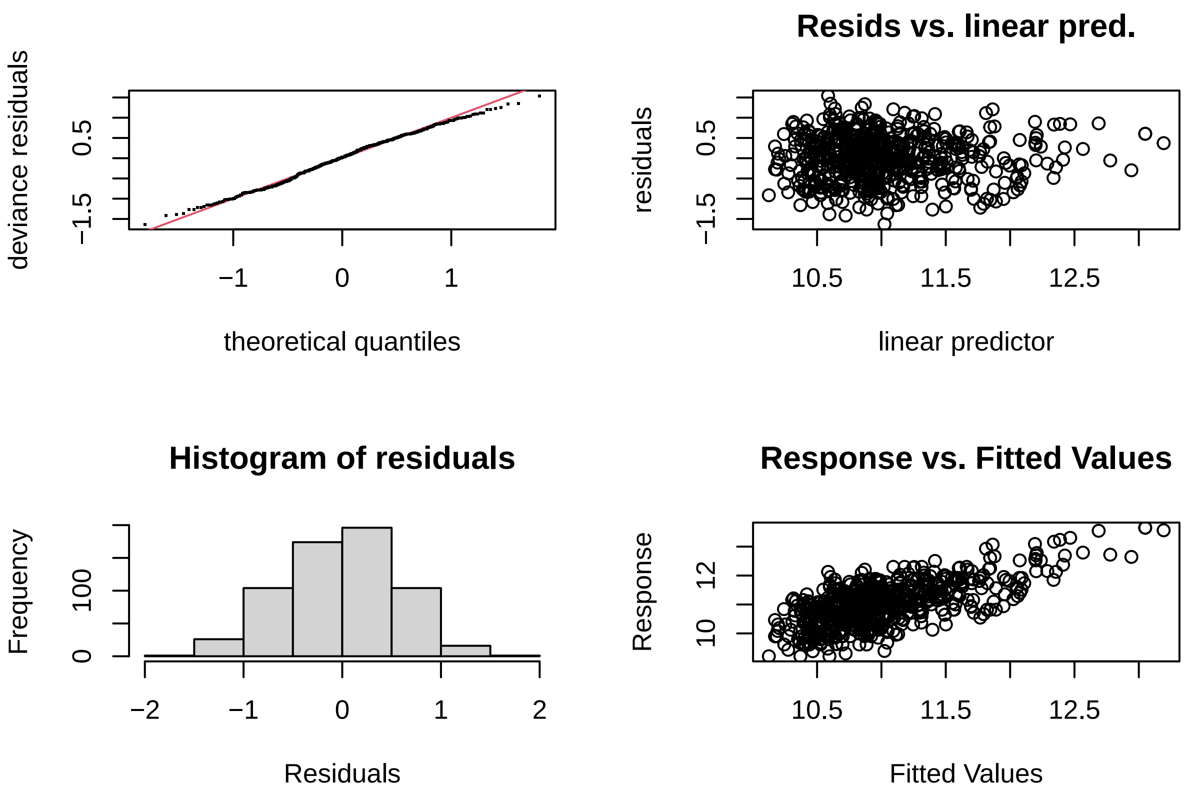
Table 1: District 6 LN_SPPSF NLM I Assessing Coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
622	0.0008536	0.0010363	0.0006689	1.549173	56.96426	-0.0007293

Model II

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_price ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##     bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + s(months,
##     bs = "cs", k = 10)
##
## Parametric coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)         10.94018    0.07340 149.048  < 2e-16 ***
## arterial             0.22896    0.08067   2.838 0.004691 **
## single_family1story  0.12482    0.06268   1.991 0.046889 *
## half_duplex1story    0.00000    0.00000     NaN      NaN
## half_duplex2story    0.07240    0.67994   0.106 0.915234
## smallest            -0.39885    0.09506  -4.196 3.14e-05 ***
## small               -0.21169    0.07209  -2.937 0.003448 **
## large                0.06854    0.07080   0.968 0.333371
## largest              0.44686    0.13297   3.361 0.000828 ***
## two_plus_baths       0.57834    0.15254   3.791 0.000165 ***
```

```
## powder_room          0.35315    0.08880   3.977 7.85e-05 ***
## good                 0.48666    0.14471   3.363 0.000821 ***
## fair                -0.27713    0.05148  -5.383 1.06e-07 ***
## poor                -0.37111    0.10916  -3.400 0.000720 ***
## very_poor_unsound   -0.28651    0.41916  -0.684 0.494542
## qabove_avg           0.54905    0.34579   1.588 0.112865
## qbelow_avg           0.02534    0.05795   0.437 0.662078
## crawl               -0.05440    0.08156  -0.667 0.505060
## slab                 0.09398    0.11440   0.822 0.411686
## fireplace           -0.22624    0.08008  -2.825 0.004888 **
## hot_water           -0.02118    0.06324  -0.335 0.737757
## elec_baseboard       0.00000    0.00000     NaN      NaN
## wall                 0.06938    0.15857   0.438 0.661855
## central_air          0.15701    0.08967   1.751 0.080469 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                                  edf Ref.df     F  p-value
## s(ln_landratio)            0.0002075      9 0.000   0.6103
## s(ln_base_bldgsize_ratio) 0.0003549      9 0.000   0.4517
## s(sresb_yearbuilt)         4.4081168      9 5.382  < 2e-16 ***
## s(ln_garagespaces)         1.1596649      3 1.273   0.0447 *
## s(months)                  2.3939667      9 3.077 6.66e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Rank: 61/63
## R-sq.(adj) =  0.422   Deviance explained = 44.9%
## -REML = 571.29  Scale est. = 0.3295    n = 622
```



```
##
## Method: REML   Optimizer: outer newton
## full convergence after 10 iterations.
## Gradient range [-7.514293e-05,0.000143223]
## (score 571.2877 & scale 0.3295034).
## Hessian positive definite, eigenvalue range [6.248932e-05,299.0221].
## Model rank =  61 / 63
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                                k'      edf k-index p-value
## s(ln_landratio)           9.000000 0.000208    0.96   0.145
## s(ln_base_bldgsize_ratio) 9.000000 0.000355    0.94   0.065 .
## s(sresb_yearbuilt)        9.000000 4.408117    0.92   0.010 **
## s(ln_garagespaces)        3.000000 1.159665    0.90   0.025 *
## s(months)                 9.000000 2.393967    0.90   0.005 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1   622        0.693      0.838       0.606  1.38  53.6 -0.584
```

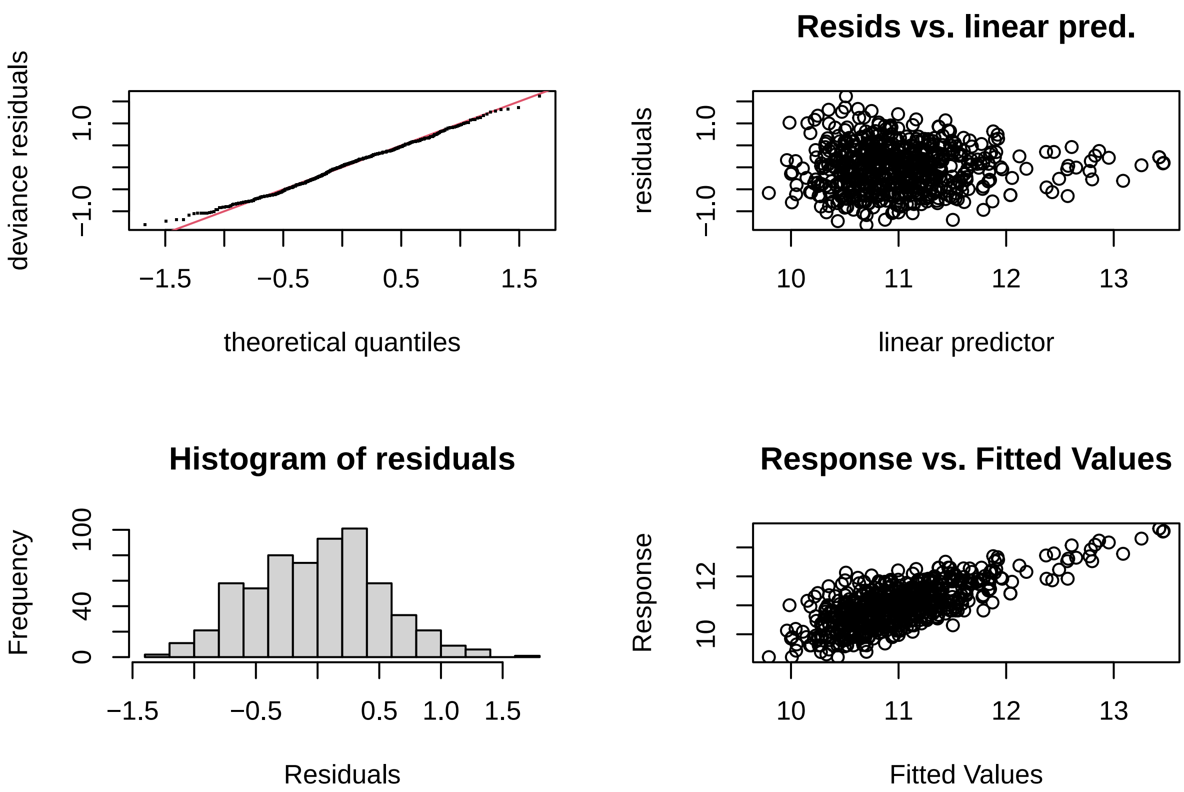
Table 2: District 6 LN_PRICE (No ECFs) NLM I Assessing Coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
622	0.9832243	1.171548	0.8486172	1.380538	52.15011	-0.6598155

Model III

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_price ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##     bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + s(months,
##     bs = "cs", k = 10) + ecf
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         10.371820   0.135058  76.795  < 2e-16 ***
## arterial             0.169319   0.076673   2.208 0.027614 *
## single_family1story  0.087749   0.058985   1.488 0.137386
```

```
## half_duplex1story    0.000000   0.000000     NaN      NaN
## half_duplex2story   -0.289138   0.694684  -0.416 0.677407
## smallest            -0.356648   0.088695  -4.021 6.56e-05 ***
## small               -0.141261   0.068473  -2.063 0.039556 *
## large                0.117116   0.066201   1.769 0.077406 .
## largest              0.357184   0.127786   2.795 0.005359 **
## two_plus_baths       0.152702   0.154057   0.991 0.321997
## powder_room          0.197864   0.086078   2.299 0.021881 *
## good                 0.333416   0.146514   2.276 0.023231 *
## fair                -0.246096   0.048858  -5.037 6.33e-07 ***
## poor                -0.292862   0.102111  -2.868 0.004280 **
## very_poor_unsound   -0.075271   0.393978  -0.191 0.848550
## qabove_avg           0.577244   0.331721   1.740 0.082364 .
## qbelow_avg          -0.010584   0.060775  -0.174 0.861808
## crawl               -0.072946   0.077320  -0.943 0.345855
## slab                 0.005863   0.108544   0.054 0.956944
## fireplace           -0.031518   0.086978  -0.362 0.717207
## hot_water           -0.053964   0.060948  -0.885 0.376303
## elec_baseboard       0.000000   0.000000     NaN      NaN
## wall                -0.146885   0.149406  -0.983 0.325957
## central_air          0.104810   0.083428   1.256 0.209513
## ecf6R602             0.251941   0.142733   1.765 0.078071 .
## ecf6R603             0.229970   0.134747   1.707 0.088418 .
## ecf6R604             0.541178   0.136150   3.975 7.93e-05 ***
## ecf6R605             0.750153   0.148396   5.055 5.78e-07 ***
## ecf6R606             0.553878   0.176983   3.130 0.001839 **
## ecf6R607             1.101013   0.179544   6.132 1.61e-09 ***
## ecf6R608             1.909372   0.239480   7.973 8.31e-15 ***
## ecf6R609             2.284039   0.341896   6.681 5.60e-11 ***
## ecf6R611             0.830628   0.138229   6.009 3.30e-09 ***
## ecf6R612             0.843275   0.175539   4.804 1.98e-06 ***
## ecf6R613             1.097017   0.159644   6.872 1.65e-11 ***
## ecf6R614             1.852904   0.294146   6.299 5.92e-10 ***
## ecf6R615             0.848566   0.152691   5.557 4.18e-08 ***
## ecf6R616             1.190843   0.297251   4.006 6.98e-05 ***
## ecf6R617             1.242091   0.340464   3.648 0.000288 ***
## ecf6R618             0.499787   0.203696   2.454 0.014438 *
## ecf6R620             0.417286   0.132404   3.152 0.001708 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                                edf Ref.df     F  p-value
## s(ln_landratio)          0.3422205      9 0.056   0.2146
## s(ln_base_bldgsize_ratio) 0.0007103      9 0.000   0.6075
## s(sresb_yearbuilt)       1.4490503      9 0.531   0.0354 *
## s(ln_garagespaces)       0.6304573      3 0.441   0.1318
## s(months)                1.9639455      9 2.595 7.05e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Rank: 78/80
## R-sq.(adj) =  0.506   Deviance explained =   54%
## -REML =  523.4  Scale est. = 0.2813    n = 622
```




```
##
## Method: REML   Optimizer: outer newton
## full convergence after 13 iterations.
## Gradient range [-0.0002193705,0.0001359973]
## (score 523.4029 & scale 0.2813027).
## Hessian positive definite, eigenvalue range [0.0002192629,290.5054].
## Model rank =  78 / 80
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                                k'     edf k-index p-value
## s(ln_landratio)           9.00000 0.34222    1.03   0.740
## s(ln_base_bldgsize_ratio) 9.00000 0.00071    0.97   0.220
## s(sresb_yearbuilt)        9.00000 1.44905    1.00   0.490
## s(ln_garagespaces)        3.00000 0.63046    1.02   0.575
## s(months)                 9.00000 1.96395    0.92   0.015 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

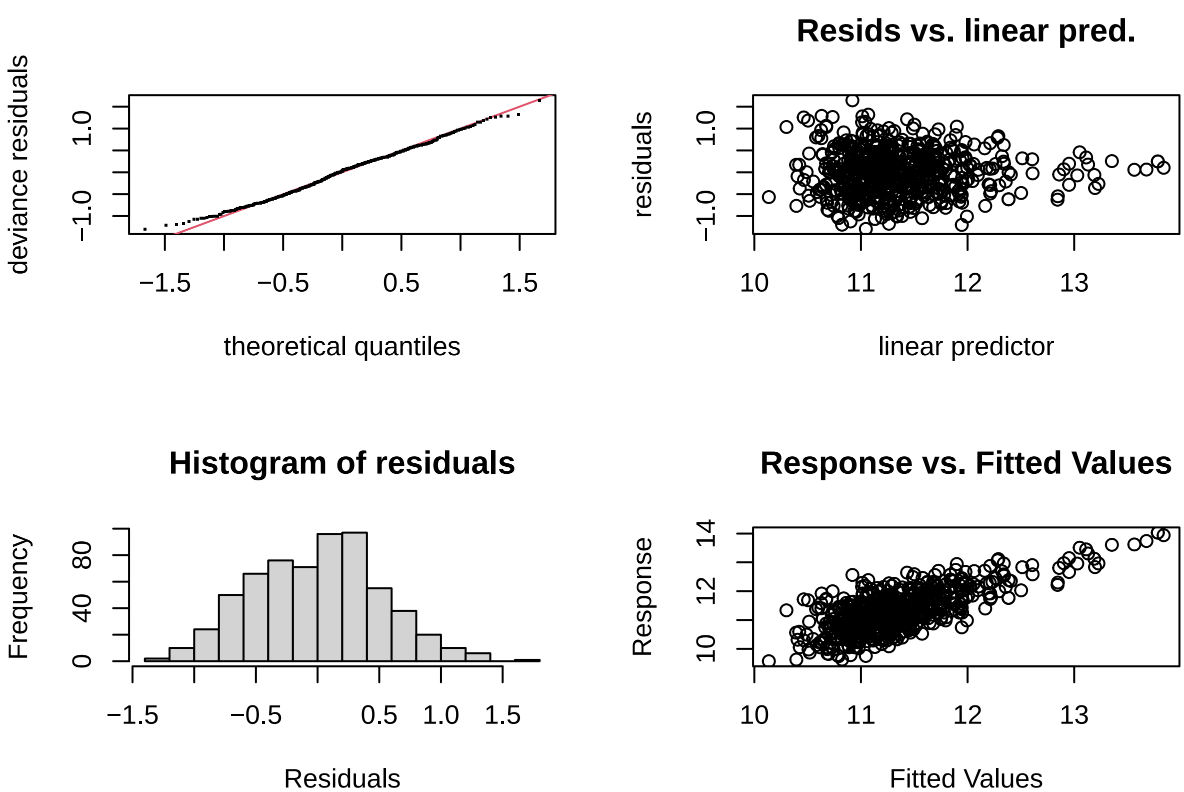
```
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1   622        0.690      0.813       0.638  1.28  48.0 -0.462
```

Model IV

```
##
```

```
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_tasp ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##     bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + ecf + s(months,
##     bs = "cs", k = 10)
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         10.704171   0.134488  79.592  < 2e-16 ***
## arterial             0.168839   0.076465   2.208 0.027631 *
## single_family1story  0.114179   0.055545   2.056 0.040268 *
## half_duplex1story    0.000000   0.000000     NaN      NaN
## half_duplex2story   -0.266851   0.691538  -0.386 0.699727
## smallest            -0.355238   0.088327  -4.022 6.54e-05 ***
## small               -0.147682   0.068088  -2.169 0.030487 *
## large                0.120013   0.065957   1.820 0.069339 .
## largest              0.372365   0.127054   2.931 0.003514 **
## two_plus_baths       0.158670   0.153507   1.034 0.301737
## powder_room          0.201329   0.085778   2.347 0.019256 *
## good                 0.330476   0.146012   2.263 0.023982 *
## fair                -0.250270   0.048656  -5.144 3.69e-07 ***
## poor                -0.294204   0.101821  -2.889 0.004003 **
## very_poor_unsound   -0.095268   0.392438  -0.243 0.808278
## qabove_avg           0.564391   0.330732   1.706 0.088451 .
## qbelow_avg          -0.013095   0.060546  -0.216 0.828844
## crawl               -0.078123   0.076673  -1.019 0.308673
## slab                 0.005975   0.108211   0.055 0.955987
## fireplace           -0.036150   0.086592  -0.417 0.676485
## hot_water           -0.052268   0.060708  -0.861 0.389608
## elec_baseboard       0.000000   0.000000     NaN      NaN
## wall                -0.139755   0.148966  -0.938 0.348548
## central_air          0.108414   0.083052   1.305 0.192284
## ecf6R602             0.245250   0.142288   1.724 0.085311 .
## ecf6R603             0.231943   0.134403   1.726 0.084927 .
## ecf6R604             0.530145   0.135602   3.910 0.000103 ***
## ecf6R605             0.723249   0.147290   4.910 1.18e-06 ***
## ecf6R606             0.560460   0.176478   3.176 0.001573 **
## ecf6R607             1.093937   0.178842   6.117 1.76e-09 ***
## ecf6R608             1.902918   0.238762   7.970 8.45e-15 ***
## ecf6R609             2.265847   0.340809   6.648 6.84e-11 ***
## ecf6R611             0.823261   0.137712   5.978 3.94e-09 ***
## ecf6R612             0.837202   0.174429   4.800 2.02e-06 ***
## ecf6R613             1.093319   0.159253   6.865 1.71e-11 ***
## ecf6R614             1.825748   0.292821   6.235 8.70e-10 ***
## ecf6R615             0.839440   0.152129   5.518 5.17e-08 ***
## ecf6R616             1.152782   0.295862   3.896 0.000109 ***
## ecf6R617             1.230233   0.339351   3.625 0.000314 ***
## ecf6R618             0.506087   0.202440   2.500 0.012696 *
## ecf6R620             0.408532   0.131763   3.101 0.002025 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Approximate significance of smooth terms:
##                                edf Ref.df     F p-value
## s(ln_landratio)          0.2674815      9 0.040  0.2387
## s(ln_base_bldgsize_ratio) 0.0002051      9 0.000  0.5983
## s(sresb_yearbuilt)        1.4729007      9 0.565  0.0305 *
## s(ln_garagespaces)        0.5884953      3 0.388  0.1456
## s(months)                 0.0002552      9 0.000  0.9218
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Rank: 78/80
## R-sq.(adj) =  0.497   Deviance explained = 52.9%
## -REML = 519.48  Scale est. = 0.27997    n = 622
```




```
##
## Method: REML   Optimizer: outer newton
## full convergence after 12 iterations.
## Gradient range [-0.0001211168,4.696793e-05]
## (score 519.4784 & scale 0.2799676).
## Hessian positive definite, eigenvalue range [6.20557e-05,290.5022].
## Model rank =  78 / 80
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                                 k'      edf k-index p-value
## s(ln_landratio)           9.000000 0.267482    1.02   0.725
## s(ln_base_bldgsize_ratio) 9.000000 0.000205    0.97   0.175
## s(sresb_yearbuilt)        9.000000 1.472901    0.99   0.380
## s(ln_garagespaces)        3.000000 0.588495    1.01   0.595
```

```
## s(months)                 9.000000 0.000255    0.92   0.025 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1   622        0.970       1.14       0.892  1.28  46.6 -0.450
```